

Ethan C. Segrue

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EDUCATION

The Ohio State University, Columbus, OH

Master of Science in Aerospace Engineering | *Expected Graduation: May 2027*

GPA: N/A

The Ohio State University, Columbus, OH

Bachelor of Science in Aerospace Engineering | *Expected Graduation: May 2026*

GPA: 3.82/4.00

QUALIFICATIONS

- **Software:** MATLAB, Python, Arduino, SolidWorks, ANSYS, Draftsight, LabVIEW, OpenVSP, and Microsoft Office
- **Skills:** Programming, CAD, data analysis, soldering, GD&T, manual & CNC machining, computer vision, 3D Printing

EXPERIENCE

Laboratory for Autonomy in Data-Driven and Complex Systems | Graduate Researcher, *May 2025 - Present*

- Use MATLAB to design a discrete-event model for alignment of asynchronous, multi-rate sensor data
- Develop an ensemble Kalman Filter to fuse infrared (IR) flight data with a predictive fire dynamics model
- Determine the intrinsic and radial distortion parameters of an IR sensor using a calibration algorithm in Python
- Perform data analysis to generate wildfire intelligence tools in preparation for submission to the IROS '26 conference

Gas Turbine Lab | Undergraduate Research Assistant, *May 2024 - April 2025*

- Supported compressor blade-tip rub and mistuning research by updating data acquisition and control systems
- Researched ceramic matrix composite (CMC) materials for resistance heaters, evaluating the potential for in-house manufacturing
- Wired and integrated a PLC control panel and RIO cabinet, soldering connectors and creating electrical diagrams in Draftsight
- Fabricated motorized gimbal and low-speed spindle assemblies using ANSYS SpaceClaim, machine tools, and Arduino
- Led a three-axis CNC machine operation by using a vector graphics editor and g-code parsing controller
- Automated test sequences, reducing manual calibration time and improving measurement repeatability across tests

PROJECTS

C-UAS Defense Study Project Lead | *August 2025 - Present*

- Direct a team of 5 students in collaboration with the Defense Innovation Unit to evaluate counter-UAS technology
- Use object-oriented programming and a Hybrid A* path-planning algorithm to model a counter-UAS kill chain in MATLAB
- Provide data-driven results using a Markov chain and multi-armed bandit framework to optimize engagement success rates

Lunar Wireless Power Transfer Satellite | *August 2025 - Present*

- Work with a team of 5 students to plan architecture concepts for lunar surface power for the NASA RASC-AL Competition
- Design a solar power satellite (SPS) that delivers continuous, uninterrupted power using wireless power transfer (WPT)
- Perform trade studies to compare commercial off-the-shelf products and evaluate WPT technology readiness level (TRL)
- Provide >1kW of power from a 100km low lunar orbit to a five square meter receiver with a 23.1% end-to-end WPT efficiency

Buckeye Space Launch Initiative (BSLI) | *August 2023 - April 2024*

- Supported avionics integration and parachute deployment, ensuring reliable data capture and mission safety
- Designed and installed a camera housing to record the team's first onboard video using SolidWorks and 3D printing

LEADERSHIP & SERVICE

Sigma Gamma Tau National Aerospace Engineering Honor Society | Chapter President, *January 2024 - Present*

- Guide a 6-member executive board to lead professional development events, graduate panels, and community outreach programs
- Coordinate partnerships with AIAA and faculty to increase chapter engagement and access to resources for students

OSU Housing and Residence Education | Resident Advisor, *August 2023 - Present*

- Foster a community of 46 residents, facilitating events that promote inclusivity and academic success
- Work with a 30-member team in administrative tasks to enforce policy and create a supportive living environment

Buckeye STEM Bridge Program | Mentor, *June 2025 - July 2025*

- Mentored a high school student in a summer research project in the Ohio State Flight Vehicle Design and Testing Group
- Guided the use of open-source aircraft modeling software like OpenVSP to compare delta, double delta, diamond, and ogive wing aerodynamics, examining high angle-of-attack lift vortices to optimize aerodynamic performance